

IT2226 – AI INNOVATION AND DEEP LEARNING PROJECT

Project Guidebook

Version 1.1

2026 S2

School of Information Technology

1. Introduction to Learning Unit (LU)

This project LU allows students to work on an Artificial Intelligence (AI) project. You will make use of design thinking and its application in the problem-solving process for collecting and processing data, selecting and evaluating AI models, developing Gen AI capabilities and finally deploying and integrating them with the application you have developed.

Working in a team, you will apply the knowledge and skills that you have learned in the other AI LUs in this specialisation to solve a real-world problem.

This guidebook should be read together with the IT2226 Project Development & Resource Planning Guide, which covers how to plan people-hours, data, compute and tooling for the two development areas in this project (ML/DL training, and Gen AI), and where to find datasets, platforms and frameworks.

Read Section 7 before you start building. This project is assessed on what you can demonstrate about your solution, not only on the solution itself. When AI coding assistants are used and what is the expectation — the marks come from evidence of your own capability. Section 7 explains exactly how that evidence is gathered.

2. Learning Outcomes

On the successful completion of this LU, you will be able to:

1. Describe the challenges and constraints faced in applying AI solutions.
2. Describe the use of AI-enabled cloud services used in the enterprise.
3. Apply the appropriate machine learning tools and platforms to build AI applications that solve real world problems.
4. Plan and execute a project as a team iteratively and deliver the solutions in a timely manner according to the given requirements.
5. Justify, verify and take responsibility for AI-assisted development work, including work produced with the help of code-generation tools.

3. Project Organisation

Project teams will consist of 3 to 4 students. Each team must elect a team leader.

Role of Team Leaders

- Coordinate and assign work among members.
- Ensure that the team is complying with project requirements stated in the project guidebook.
- Keep track of the project schedule and ensure that the scheduled tasks are completed on time.

- Arrange meetings with project supervisor; this includes the booking of meeting venues.
- Communicate with the supervisor and external party on behalf of the team as and when necessary.
- Ensure every member commits their own work to the team repository under their own account (see Section 7.4).

Role of Team Members

- Participate actively in all activities during practical and tutorial.
- Cooperate with the rest of the team and complete all the tasks allocated. (Marks may be deducted for individuals who rely heavily on others to complete his/her tasks. This is to cultivate an independent working attitude.)
- Be present and punctual for all group meetings.
- Respect the authority and decisions of the team leader.
- Take personal responsibility for every line of code submitted under their name, including code produced with AI assistance.

Project Supervisors (LU Tutor)

- Provide feedback on project teams' proposals and designs.
- Advise the project team during project development, on matters including task allocation, tools used, etc.
- Provide inputs from the company management / user perspective.
- Conduct the individual code interrogation and live task at each review checkpoint (Section 7.2).

4. Project Details

Develop an ethical AI application that solves a real problem to benefit the lives of Singaporeans, under one of the following themes:

- Business
- Education
- Healthcare
- Logistics
- Smart Nation
- Sustainability
- Transport

Teams can research into problem statements surfaced in recent or current SG challenges, hackathons or competitions and choose/adapt one to work on. Using Design Thinking processes, you will then seek to understand your target users, discover what they would like to see in your solution, brainstorm solutions in line with the theme before proposing a solution that fits the theme. As a team, you will subsequently apply what you have learned in this or other learning units to collect and process data, train AI supervised learning models, develop Gen AI capabilities and integrate them into the application to showcase your proposed solution.

Section 4 of the ***Resource Planning Guide*** lists starting points for researching problem statements (Deloitte AI Dossier, LeewayHertz, data.gov.sg, LTA DataMall, Smart Nation Singapore).

5. Scope of Work

Each team member is required to:

1. Collect and understand the data and ensure that the data is prepared in a way that is readily understood by the Machine Learning (ML) model.
2. Train a classical or deep supervised learning model, conduct hyperparameter tuning and use suitable metrics (such as accuracy, precision, recall) to evaluate the performance.
3. Document the data collection, understanding, preparation and model training experiments.
4. Develop a Gen AI capability with clearly defined use cases. This may be a RAG capability, a fine-tuned model, or an agentic (single- or multi-agent) capability built with a framework such as LangGraph or CrewAI.
5. Develop an application that has your ML model and Gen AI capability integrated.
6. Develop additional features (e.g. user management, interface to API services like Maps API, mobile app, etc.) to enhance your team solution.
7. Present your work in detail, and be able to explain and modify any part of it on request.

6. Tasks & Schedule

Week	Tasks	Individual (%)	Group (%)
2	• Administrative Tasks • Finalise teams • Draft problem statement • Set up team repository — every member commits under their own account • Start the AI Use Log (Annex B)	–	–
3	• Idea Presentation • Technical Spec (Individual) • Slides (Group)	-	-
5	• Progress Review 1 • Supervised learning model • Code Interrogation + Live Task • AI Use Log • Team Progress	15	5
14	• Progress Review 2 • Gen AI / agentic capability • Code Interrogation + Live Debug • AI Use Log	25	5
18	• Final Presentation • Final Report • Slides • Source Codes + version history • Individual Technical Defence • AI Use Log (final) • Peer Review • Ethical AI Checklist	35	5

Week	Tasks	Individual (%)	Group (%)
Total		**75**	**15**

1)Individual marks total 75%; group marks total 15%.

2)Mock Interview 10%

The overall LU total is 100%.

7. How Your Work Is Assessed in an AI-Assisted Environment

AI coding assistants can now produce a working, well-structured, well-commented application from a paragraph of description. That is a legitimate and valuable way to work — you are expected to use these tools, and you will be at a disadvantage in industry if you do not. It does, however, mean that a polished artefact on its own no longer tells your tutor anything about what you can do.

So this LU assesses the artefact and the author. Roughly 30% of the LU total is carried by criteria that can only be earned in person, on your own build: explaining it, changing it, and debugging it while your tutor watches. Nothing you cannot explain or modify counts as evidence of your capability, however well it works.

7.1 Three channels of evidence

Channel	What it is	What it is good for
The artefact	The working model, capability and application you submit.	Shows the solution is real and functions. Necessary, but cheap to produce — so it carries fewer marks than before.
The process record	Version-control history and the AI Use Log (Annex B).	Shows the work happened over time, by you, with iteration, corrections and abandoned attempts.
The live defence	Unprepared code interrogation, a tutor-set modification, and a seeded-fault debug at each checkpoint.	Shows understanding and capability directly. This is where the discriminating marks are.

7.2 What happens at a checkpoint

At each of Progress Review 1, Progress Review 2 and the Final Presentation, every member individually goes through a short session with the tutor(might not evaluate on the fixed date). Notes, prepared scripts and AI assistants are closed for this part. Expect roughly:

- Code walkthrough — open a file you are credited with and explain what it does, why it is there, what depends on it, and what would break if it were removed. Expect follow-up questions on the parts you did not choose to talk about.
- Live task — a small change set by the tutor: add a field, change a metric, handle an additional input case. You may use your normal editor. What is marked is whether you know where to change it, whether you verify the result, and whether you anticipate the side effects.

- Seeded-fault debug — the tutor introduces a defect into a copy of your code. What is marked is your method: reproduce, isolate, hypothesise, test — not the speed of the fix.
- AI practice — walk through two or three entries in your AI Use Log: what you asked for, what came back, what you changed and why.

7.3 What earns high marks

The top band is not "did not use AI". It is skilled, verified, well-directed use. Specifically, the A-band descriptors reward students who:

- Direct the tool with precise specifications rather than vague prompts.
- Verify output against tests, data or documentation instead of against "it ran without an error".
- Catch and correct substantive errors — a hallucinated API, a metric computed on the wrong split, data leakage, a dataset link that does not exist.
- Reject or rewrite generated code where it is wrong for this problem, and can say why.
- Can name the parts of their own codebase they would not trust in production.

Conversely, unverified generated code in the critical path, or an AI Use Log that cannot be reconciled with the version history, places the relevant criterion in the Inadequate band regardless of whether the application runs.

7.4 Version control and individual contribution

- Set up the team repository in Week 2. Every member commits from their own account.
- Commit small and often. A project with three commits, all from one account, cannot evidence individual contribution — and the Individual Contribution Evidence criterion at the Final will be marked accordingly.
- Bulk imports of generated code should be committed separately from your own edits, so the difference is visible.
- The version history is evidence, not admin. Treat it as part of the submission.

7.5 Academic integrity

Using an AI assistant is not plagiarism in this LU. Submitting work you cannot account for is. Two things are treated as academic misconduct:

- Declaring in the AI Use Log work that is not what you actually did, or submitting a log that contradicts the version history.
- Presenting another team's or another person's work — human or generated — as your own individual contribution.

The applicable AI tool use policy for this run of the LU is stated in Annex C and will be confirmed by your tutor in Week 2.

8. Project Supervisor

LU Leader & Tutor	Mr. Wee Chee Hong
Contact	[Phone – 65501754] [Email – wee_chee_hong@nyp.edu.sg]

Annex A – Suggested Content

1. Presentation
    a. Project Plan (Individual)
        - Scope of work
            - Supervised Learning Model
            - Gen AI Capability
        - Milestones & Deliverables
        - Reference links (for any research done)
    b. Slides (Group)
        - Problem statement
        - Research into problem
        - Individual component
        - Schedule

2. Progress Review/Final Report
    a. Solution
    b. Models
        - Supervised Learning Model
        - Gen AI Capability
    c. Features
    d. UX
    e. Reflection

Annex B – AI Use Log

Every member keeps their own log, updated as you work rather than reconstructed at the end. Submit it at each review checkpoint. It does not need to record every autocomplete — log anything that produced a block of code, a design decision, or a resource you then used.

Date	Tool	What you asked for	Accepted / changed / rejected	Why — what you checked and what you found
12 Feb	Claude Code	Generate the FastAPI endpoint that wraps the classifier	Changed — rewrote the input validation	Generated code accepted any JSON body; added a Pydantic schema so bad requests fail at the boundary instead of inside the model call
14 Feb	Copilot	Suggest an evaluation block for the triage classifier	Rejected	Computed accuracy on the training split. Rewrote to evaluate on the held-out test set — the original number was meaningless
19 Feb	ChatGPT	Find a public dataset for symptom-to-urgency labels	Rejected 2 of 3 suggestions	Two of the three Kaggle links returned 404. Used the third after downloading and checking the label distribution

The "Why" column is the one that is marked. A log full of "accepted – it worked" evidences nothing. The examples above are the standard: a specific check, a specific finding, a specific action.

Rubrics

1. Progress Review 1 — Week 5 (20%)

Individual					
Criteria	Inadequate (F)	Acceptable (D)	Satisfactory (C)	Good (B)	Excellent (A)
Input Data [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Collection	• No data or irrelevant data collected	• Data collected is relevant but only sufficient for use with pre-trained models	• Data collected is relevant and marginally sufficient from a single source	• Data collected is relevant and substantial from a single source	• Data collected is relevant, substantial and coherent; process is sophisticated
EDA/Preparation	• No EDA done	• Minimum EDA done and many issues with data preparation	• Moderate EDA done and some issues with data preparation	• Good EDA with simple explanations and no issue with data preparation	• Excellent EDA with insightful details, use of visualisation resulting in comprehensive data preparation
Supervised Model [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Training	• None or incomplete	• Completed using only pre-trained models without subsequent training	• Completed with custom training and major conceptual errors	• Completed with custom training and minor conceptual errors	• Completed with custom training; no conceptual errors and evidence of optimisations
Evaluation	• None or inappropriate use and interpretation of performance metrics	• Appropriate use and interpretation of performance metrics with major misconceptions	• Appropriate use and interpretation of performance metrics with moderate misconceptions	• Appropriate use and interpretation of performance metrics with minor misconceptions	• Appropriate use and in-depth interpretation of performance metrics with no misconceptions
Source Codes [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Structure	• None and no comments or markdown for explanations	• Poorly organised with confusing flows and poor use of comments or markdown	• Organised with logical flows with sparse or poor use of comments or markdown	• Organised with logical flows and good use of comments or markdown	• Well-organised, logical, modular with excellent use of comments or markdown

Code Review	• Unable to explain critical components of the codes or provide clarifications when probed	• Able to explain limited critical component of the codes conveying sparse subject knowledge	• Able to explain some critical components of the codes conveying decent subject knowledge	• Able to explain most critical components of the codes conveying good subject knowledge	• Able to explain all critical components of the codes conveying excellent subject knowledge
Group					
Plan and Presentation [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Progress	• No details on refining plan to address new requirements or changes in scope	• Poor progress update with significant gaps in revised plan to address changes in scope	• Some progress update with minor gaps in revised plan to address changes in scope	• Good progress update with minimal gaps in revised plan to address changes in scope	• Comprehensive progress update with no gaps and sound recommendations in revised plan to address changes in scope
Slides	• None or irrelevant	• Content is relevant but fragmented and poorly organised	• Content is relevant and moderately organised	• Content is relevant, well organised and coherent	• Content is relevant, well organised, coherent and extensive
Execution	• Complete lack of preparation resulting in incoherent and confusing execution	• Poor preparation between members resulting in major execution problems	• Moderate preparation between members resulting in minor execution problems	• Good preparation between members resulting in minimal execution problems	• Excellent preparation between members resulting in excellent execution

2. Progress Review 2 — Week 14 (30%)

Individual					
Criteria	Inadequate (F)	Acceptable (D)	Satisfactory (C)	Good (B)	Excellent (A)
Supervised Model [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Optimisation	• None or poor techniques resulting in accuracy < 50%	• Basic techniques resulting in accuracy ≥ 50%	• Intermediate techniques resulting in accuracy ≥ 60%	• Good techniques resulting in accuracy ≥ 70% (or industry benchmark)	• Systematic and iterative techniques resulting in accuracy ≥ 80% (or industry benchmark)
Deployment/Hosting	• No model deployed	• Model is deployed locally with major issues	• Model is deployed locally with minor issues	• Model is deployed locally without issues	• Model is deployed remotely without issues
Gen AI Capability [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Use case and scope of work	• Unclear objectives with poorly defined scope or no avenues to achieve them	• Clear objectives with basic scope of work and research	• Clear objectives with intermediate scope of work and research	• Clear objectives with good scope of work and research	• Clear objectives with excellent scope of work and extensive research
Implementation	• None	• Achieved with major errors or implementation gaps	• Achieved with moderate errors or implementation gaps	• Achieved with minor errors or implementation gaps	• Fully achieved with no errors or gaps, complete with detailed explanations
Application [10 Marks]	Marks: 0.0 to < 5	Marks: 5 to < 6	Marks:6 to < 7	Marks: 7 to < 8	Marks: 8 to 10
Functionality	• Models not integrated leading to no results • No User Interface (UI) • No output	• Can showcase either supervised model or Gen AI capability • UI is totally not intuitive • Outputs do not illustrate problems are solved	• Can showcase both supervised model and Gen AI capability with moderate issues • UI is moderately intuitive • Outputs somewhat illustrate problems are solved	• Can showcase both supervised model and Gen AI capability with minor issues • UI is largely intuitive • Outputs largely illustrate problems are solved	• Can showcase both supervised model and Gen AI capability with no issues • UI is very intuitive • Outputs clearly illustrate that problems are solved
Source Codes [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0

Structure	• None and no comments or markdown for explanations	• Poorly organised with confusing flows and sparse use of comments or markdown	• Moderately organised with logical flows and moderate use of comments or markdown	• Largely organised with logical flows and good use of comments or markdown	• Well-organised, logical, modular with excellent use of comments or markdown
Code Review	• Unable to explain or clarify critical components of the codes when probed	• Able to explain minimum critical components of the codes conveying sparse subject knowledge	• Able to explain some critical components of the codes conveying moderate subject knowledge	• Able to explain most critical components of the codes conveying good subject knowledge	• Able to explain all critical components of the codes conveying excellent subject knowledge
Group					
Presentation [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Slides	• None or irrelevant	• Content is relevant but fragmented and poorly organised	• Content is relevant and moderately organised	• Content is relevant, well organised and coherent	• Content is relevant, well organised, coherent and extensive
Execution	• Complete lack of preparation resulting in incoherent and confusing execution	• Poor preparation between members resulting in major execution problems	• Moderate preparation between members resulting in minor execution problems	• Good preparation between members resulting in minimal execution problems	• Excellent preparation between members resulting in excellent execution

3. Final Project Presentation — Week 19 (40%)

Individual					
Criteria	Inadequate (F)	Acceptable (D)	Satisfactory (C)	Good (B)	Excellent (A)
Presentation [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Delivery	• Voice is consistently too weak or too strong • Fails to use inflections to emphasise key points or to create interests • Appears very uneasy and insecure • Lacks or has body motions or gestures that are highly distracting	• Voice is frequently too weak or too strong • Rarely uses inflections to emphasise key points or to create interests • Appears uneasy and somewhat insecure • Uses few body motions or gestures that distract audience	• Voice is generally strong • Sometimes uses inflections to emphasise key points • Appears generally at ease and confident • Body motions and gestures are neutral	• Voice is strong • Often uses inflections to emphasise key points • Appears mostly comfortable and confident • Uses body motions and gestures well	• Voice is confident and strong • Consistently uses inflections to emphasise key points • Appears very comfortable and confident • Uses body motions and gestures very effectively
Q & A	• Unable to respond to questions or provide irrelevant answers	• Able to respond with somewhat relevant, direct answers	• Able to respond with relevant, direct answers	• Able to respond with relevant answers and short elaborations	• Able to respond with relevant answers and insightful elaborations
App Demo [10 Marks]	Marks: 0 to < 5	Marks: 5 to < 6	Marks: 6 to < 7	Marks: 7 to < 8	Marks: 8 to 10
Model, Gen AI Capability & Features	• Models not integrated into solution or unable to showcase features due to errors or failures	• Models integrated into solution and able to showcase minimum features	• Models integrated into solution and able to showcase features with following attribute: (i) relevance	• Models integrated into solution and able to showcase features with following attributes: (i) relevance (ii) tech proficiency	• Models integrated into solution and able to showcase features with following attributes (i) relevance (ii) tech proficiency (iii) impact
User Experience (UX)	• Poor usability without regards to any UX principles or human-centric design	• Limited UX principles, low human-centric design and usability	• Basic UX principles, human-centric design and usability	• Moderate UX principles, human-centric design and usability	• Excellent UX principles, human-centric design and usability across solution
Ethical AI [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0

Implementation Guidelines	• Submitted relevant answers for < 50%	• Submitted relevant answers for ≥ 50%	• Submitted relevant answers ≥ 60%	• Submitted relevant answers ≥ 75%	• Submitted relevant answers ≥ 90%
Reflection [10 Marks]	Marks: 0 to < 5	Marks: 5 to < 6	Marks: 6 to < 7	Marks: 7 to < 8	Marks: 8 to 10
Lessons Learned	• No analysis with reference to project	• Poor analysis with reference to project	• Brief analysis involving narrow aspects of project	• Good analysis involving narrow aspects of the project	• In-depth analysis involving broad aspects of the project
Improvements	• No suggestions to improve the solution	• Suggestions to improve solution are trivial	• Suggestions to improve solution are relevant but lack details	• Suggestions to improve solution are relevant with some details	• Suggestions to improve solution are relevant, well-crafted and feasible
Peer Review [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Submission and reviews	• Received extremely poor reviews from all members	• Members highlighted major issues	• Members highlighted moderate issues	• Members highlighted minor issues	• No issues highlighted by members
Group					
Presentation [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Slides	• None or irrelevant	• Content is relevant but fragmented and poorly organised	• Content is relevant and well organised	• Content is relevant, well organised and coherent	• Content is relevant, well organised, coherent and extensive
Execution	• Complete lack of preparation resulting in incoherent and confusing execution	• Poor preparation between members resulting in major execution problems	• Moderate preparation between members resulting in minor execution problems	• Good preparation between members resulting in minimal execution problems	• Excellent preparation between members resulting in excellent execution

End of Document